

Nonlinear Aggregation of Composite Sub-scores: The Godambe Gap and the Auxiliary-Parameter Problem

Research note — theory and algorithms

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Abstract

This note consolidates the theoretical results and the algorithms they justify. Part I characterizes when nonlinear aggregation recovers the Fisher score. A theorem pair (separable sufficiency, and its exponential-tilt converse — Pitman–Koopman–Darmois relocated to the aggregation problem) shows that the linear–nonlinear gap is *exactly* the curvature of the tilt in its separable statistic. Two distinct claims follow, with distinct load-bearing models: nonlinear beats linear for elliptical families with banded inverse scatter, with exactness iff the generator is Gaussian; and *pre-aggregation* nonlinearity beats outer nonlinearity precisely when there are many independent latent units or higher-order cliques, where a CLT-washing argument shows outer nonlinearity provably cannot substitute. A single global latent is conceded to outer nonlinearity explicitly. Part II resolves the *auxiliary-parameter problem*: the surrogate score must be evaluated at some β , and amortized SBI forces $\beta = \hat{\beta}(X)$ to be data-dependent. A single translation identity, forced by Fisher consistency, yields (i) an impossibility result — no function of the score value alone can be first-order pilot-invariant without becoming ancillary — and (ii) a complete characterization: on the pair $(\hat{\beta}, S_{\psi, \phi}(X, \hat{\beta}))$ the first-order pilot-invariant statistics are *exactly* the functions of the one-step estimator $\hat{\theta}_1$. The degeneracy of $S_{\psi, \phi}(X, \hat{\theta}_1)$, and the vanishing of the score at the estimating-equation root, are corollaries of the same identity. Part III separates *validity* from *efficiency*: in the SBI regime pilot error never threatens posterior validity, only sharpness, and the conditions the pilot must satisfy are tiered accordingly. Part IV derives the algorithms as consequences and maps each claim to an experiment.

Status flags. Results marked [PROVED] are complete as stated. Results marked [TO VERIFY] require a model-specific computation that is sketched but not carried out here. The principal [TO VERIFY] items are Theorem 14 (the blockwise gap, which carries the headline claim) and the injectivity condition in Theorem 8; both should be settled before drafting, since the framing depends on whether they are theorems or empirical findings.

1 Setup and notation

Let $X = (X_1, \dots, X_n) \sim p(\cdot \mid \theta)$ with $\theta \in \Theta \subset \mathbb{R}^p$ and let

$$g_0(X; \theta) = \nabla_{\theta} \log p(X \mid \theta)$$

denote the full Fisher score, assumed unavailable or too expensive. Fix tractable marginal and pairwise densities p_1, p_2 and define sub-scores

$$\text{cs}_1(X_i; \beta) = \nabla_{\beta} \log p_1(X_i \mid \beta), \quad \text{cs}_2(X_i, X_j; \beta) = \nabla_{\beta} \log p_2(X_i, X_j \mid \beta).$$

With $\{(i, j) : i < j\} = \bigcup_{k=1}^d U_k$ a partition into pair classes (lags, distance bins), define the aggregated features

$$\text{CS}_1(X; \beta, \phi) = \sum_{i=1}^n a_\phi \{\text{cs}_1(X_i; \beta)\}, \quad \text{CS}_{2,k}(X; \beta, \phi) = \sum_{(i,j) \in U_k} b_{\phi,k} \{\text{cs}_2(X_i, X_j; \beta)\},$$

stacked as $T_\phi(X, \beta) = \{\text{CS}_1, \text{CS}_{2,1}, \dots, \text{CS}_{2,d}\}$, and the surrogate score

$$S_{\psi,\phi}(X, \beta) = S_\psi\{T_\phi(X, \beta)\} \in \mathbb{R}^p.$$

Write the centered estimating function

$$m(X, \beta) = S_{\psi,\phi}(X, \beta) - \mathbb{E}_\beta\{S_{\psi,\phi}(X, \beta)\}, \quad \mu(\theta, \beta) = \mathbb{E}_\theta\{m(X, \beta)\},$$

so that $\mu(\theta, \theta) = 0$ for all θ by construction. Sensitivity, variability and Godambe information at θ are

$$H(\theta) = \partial_\theta \mu(\theta, \beta) \big|_{\beta=\theta}, \quad V(\theta) = \text{Var}_\theta\{m(X, \theta)\}, \quad G(\theta) = H^\top V^{-1} H.$$

The *one-step* (Newton) estimator from a pilot $\hat{\beta} = \hat{\beta}(X)$ is

$$\hat{\theta}_1 = \hat{\beta} + H(\hat{\beta})^{-1} m(X, \hat{\beta}), \tag{1}$$

and $\hat{\theta}_k$ denotes k iterations of (1) (Fisher scoring), with $\tilde{\theta}$ the exact root $m(X, \tilde{\theta}) = 0$.

Assumption 1 (Regularity). $\mu(\theta, \beta)$ is C^2 in a neighbourhood of the diagonal; $H(\theta)$ is non-singular and continuous; $V(\theta) \succ 0$; and $n^{-1/2}m(X, \theta_0) \rightsquigarrow N(0, V)$ with the usual stochastic-equicontinuity conditions for a LAN expansion.

Assumption 2 (Fisher consistency). $\mu(\theta, \theta) = 0$ for every $\theta \in \Theta$.

Assumption 2 is not an extra requirement: it holds by construction once $S_{\psi,\phi}$ is centered, and is the defining property of a valid estimating function. Everything in Part II is a consequence of differentiating it.

2 Part I: when nonlinear aggregation recovers the Fisher score

Part I answers two distinct questions that the earlier draft conflated:

- (Q1) When does *nonlinear* aggregation of sub-scores beat the best *linear* composite score? (Justifies the method against classical composite likelihood.)
- (Q2) When does *pre-aggregation* nonlinearity (ϕ applied to each local sub-score before summing) beat *outer* nonlinearity applied to linear sums? (Justifies the feature networks $a_\phi, b_{\phi,k}$ specifically.)

These have different answers, and the models that settle (Q1) mostly cannot settle (Q2). Theorem 8 answers (Q1); Theorem 14 answers (Q2); Proposition 12 states honestly where (Q2) fails.

2.1 The statistic class and the recovery criterion

Let $\mathcal{U} = \{W_i\}$ be a *locality covering* of the data — single sites, pairs, k -tuples, sliding windows, latent blocks, or replicates — and let $u_i(Y; \beta)$ be the vector of exact sub-scores computable on window W_i at the pilot β . The nCS architecture is the statistic class

$$\mathcal{N}(\mathcal{U}) = \left\{ q\left(\sum_i \phi(u_i(Y; \beta)); \beta\right) : \phi \in \mathcal{C}(\mathbb{R}^d, \mathbb{R}^m), q \in \mathcal{C}, m < \infty \right\}, \quad (2)$$

i.e. local nonlinearity, sum, outer map. Three nested sub-classes matter throughout:

$$\underbrace{\{W^\top \sum_i u_i\}}_{\text{Method A: linear CS}} \subset \underbrace{\{q(\sum_i u_i)\}}_{\text{Method C: outer nonlinearity}} \subset \underbrace{\mathcal{N}(\mathcal{U})}_{\text{nCS: pre-aggregation nonlinearity}}.$$

Method A is $\phi = \text{id}$ with q linear; Method C is $\phi = \text{id}$ with q arbitrary. The document’s $\mathcal{L}$ of Section 2.3 is the L^2 closure of Method A. Separating C from A is essential: much of what looks like “nonlinearity helps” is achieved by C alone, and C requires no feature networks.

2.2 Exact recovery and its converse

Theorem 1 (Separable sufficiency $\Rightarrow$ exact recovery; PROVED). *Suppose there exist finite m , measurable maps ψ_i , and functions g, h with*

$$(i) \quad p(y; \theta) = g\left(\sum_i \psi_i(W_i(y)); \theta\right) h(y) \quad \text{for all } \theta \text{ near } \beta, \quad (3)$$

$$(ii) \quad \psi_i(W_i) = \phi(u_i(Y; \beta)) \quad \text{a.s. } [P_\beta] \text{ for some measurable } \phi. \quad (4)$$

Condition (3) is additive separability of a sufficient statistic over the covering; (4) is local score-sufficiency. Then

$$g_0(Y; \beta) = \nabla_\theta \log g(T; \theta)|_\beta = q\left(\sum_i \phi(u_i); \beta\right) \in \mathcal{N}(\mathcal{U}),$$

and nCS attains relative efficiency 1.

Proof. Differentiate (3) in θ ; h does not depend on θ and drops out. Substitute (4). $\square$

Theorem 2 (Converse: exponential-tilt characterization; PROVED). *Suppose $g_0(\cdot; \theta) = q(T(\cdot); \theta)$ for a fixed statistic $T = \sum_i \phi(u_i)$ and all θ in a neighbourhood Θ_0 of β . Then*

$$p(y; \theta) = p(y; \beta) \exp\{\Lambda(T(y); \theta)\}, \quad \Lambda(t; \theta) = \int_\beta^\theta q(t; \vartheta)^\top d\vartheta,$$

so T is sufficient and the family is an exponential tilt of $p(\cdot; \beta)$ in T . Hence

$$\boxed{g_0(\cdot; \theta) \in \mathcal{N}(\mathcal{U}) \text{ for all } \theta \text{ near } \beta \iff p(\cdot; \theta) = p(\cdot; \beta) e^{\Lambda(\sum_i \phi(u_i); \theta)}}.$$

Proof. Fix y, y' with $T(y) = T(y')$. Then $\nabla_\theta \{\log p(y; \theta) - \log p(y'; \theta)\} = q(T(y); \theta) - q(T(y'); \theta) = 0$ on Θ_0 , so the log-likelihood difference is constant in θ ; integrate along paths from β . Sufficiency of T follows by factorization. $\square$

Remark 3 (Pitman–Koopman–Darmois, relocated). Theorem 2 is the PKD principle transplanted to the aggregation problem: exact nCS recovery *uniformly in θ* is possible precisely for (generally curved) exponential-type families whose natural statistics are sums of local-window functionals — Gibbs / Hammersley–Clifford structure with potentials supported on $\mathcal{U}$. At a *single* pilot the requirement weakens to “ $g_0(\cdot; \beta)$ is $\sigma(T)$ -measurable”, which is what regression- R^2 diagnostics measure and what the $L_0^2(p_{\theta_0})$ projection argument below uses. The two statements should not be conflated: Part I’s ceiling results are single-pilot; Theorem 2 is the stronger uniform statement.

2.3 The linear ceiling and the exact linear–nonlinear dichotomy

A *linear* composite score is

$$S_{\text{lin}}(X; \beta, W) = W_1 \sum_i \text{cs}_1(X_i; \beta) + \sum_{k=1}^d W_{2,k} \sum_{(i,j) \in U_k} \text{cs}_2(X_i, X_j; \beta). \quad (5)$$

The gap claim must be stated against the largest linear class, so allow *per-component* (Lindsay-type) weights:

$$\mathcal{L} = \overline{\text{span}} \left\{ v^\top \text{cs}_1(X_i; \theta_0), v^\top \text{cs}_2(X_i, X_j; \theta_0) : v \in \mathbb{R}^p, i < j \right\} \subset L_0^2(p_{\theta_0}),$$

with $\Pi_{\mathcal{L}}$ the L^2 projection. Under stationarity with lag-defined batches the optimal per-pair weights are constant within each lag class by exchangeability, so (5) attains the same ceiling; in general (5) is a sub-class.

Proposition 4 (Linear ceiling; PROVED). *For any W , $G(S_{\text{lin}}(\cdot; W)) \preceq G_\star^{\text{lin}} := \sup_W G(S_{\text{lin}}(\cdot; W))$, and*

$$G_\star^{\text{lin}} = \mathbb{E}[\Pi_{\mathcal{L}} g_0 (\Pi_{\mathcal{L}} g_0)^\top] = I(\theta_0) - \mathbb{E}[RR^\top], \quad R := g_0 - \Pi_{\mathcal{L}} g_0.$$

Hence $G_\star^{\text{lin}} = I(\theta_0)$ if and only if $g_0 \in \mathcal{L}$.

Proof sketch. Godambe information is invariant to nonsingular linear reparametrization and is maximized over a linear class by the projection of the efficient score onto that class [Godambe, 1960, Heyde, 1997]; the Pythagorean identity in L_0^2 gives the residual decomposition. The optimal $W^\star$ corresponds to $M^\star = V^{-1}H$ in $\text{tr}(G) = \max_M \text{tr}(2M^\top H - M^\top V M)$. $\square$

Combining Proposition 4 with Theorem 2 gives the statement that should organize the paper. Both classes are exponential tilts in a separable statistic; they differ only in the *shape* of the tilt.

Corollary 5 (Linear–nonlinear dichotomy; PROVED). *Let $T = \sum_i \phi(u_i)$ be separable-sufficient as in Theorem 1. Then*

$$\text{linear CS exact} \iff \Lambda(t; \theta) \text{ is } \mathbf{affine} \text{ in } t, \quad \text{nCS exact} \iff \Lambda(t; \theta) \text{ is an } \mathbf{arbitrary} \text{ tilt in } t.$$

The linear–nonlinear efficiency gap is therefore exactly the curvature of the tilt Λ in its statistic argument. Genuine exponential families with separable natural statistics are the fixed point at which the gap vanishes.

Proof. If Λ is affine, $g_0 = \nabla_\theta \Lambda$ is a fixed linear functional of T ; when $\phi = \text{id}$ is admissible (Level 0 below) this places $g_0 \in \mathcal{L}$. Conversely if $g_0 \in \mathcal{L}$ then g_0 is linear in the sub-score sums for every θ near β , so $\nabla_\theta \Lambda$ is t -linear and Λ is t -affine. $\square$

Proposition 6 (Level 0: separable log-likelihoods; PROVED). *If $\log p(y; \theta) = \sum_i \ell_i(W_i; \theta)$ is itself additively separable over the covering, then the tilt is affine and $\phi = \text{id}$ already suffices: **linear CS is exact**. This covers*

- (a) *i.i.d. models with $\mathcal{U} = \text{single sites}$;*
- (b) *any stationary order- k Markov model with $\mathcal{U} = \text{consecutive } (k+1)\text{-tuples}$, via $\log p = \sum_t \log f_{k+1} - \sum_t \log f_k$, provided the library contains both the $(k+1)$ - and k -dimensional margins;*

Level	Structure of $\log p$	Minimal exact recoverer	Instance
0	$\sum_i \ell_i(W_i; \theta)$	linear CS	i.i.d.; order- k Markov with $(k+1)$ -ma
1	$\Lambda(\sum_i \psi(W_i; \theta), W_i \text{ inside library})$	nCS at library granularity	elliptical banded precision; global mi
2	as Level 1, W_i larger than library	windowed/blockwise nCS	block latents; replicates; delay- k AR
3	no finite separable T (matrix product)	none exact; geometric window law	HMM; stochastic volatility; Hawkes

Table 1: The four-level ladder. The gap is governed by *where the nonlinearity sits relative to the sub-score library’s granularity*, not by “non-Gaussianity” as such.

(c) *variance-component and anisotropy models whose log-likelihood is a sum of site-wise terms.*

Remark 7 (The Level-0 trap, and what it forbids as a testbed). Proposition 6(b) is a strong and easily overlooked negative result: *every first-order Markov model has zero gap once the library contains pairwise and marginal sub-scores* — verified numerically to 5×10^{-10} on a Markov copula chain and to $R^2 = 1.000000$ on a triple-margin threshold AR. Consequences for the experiment plan:

- “Student- t AR(1)” is ambiguous and one reading is fatal. *Elliptical with AR(1) scatter* (Example 1) is Level 1 and has a gap; *AR(1) with i.i.d. t innovations* is Markov, hence Level 0, and has **zero** gap. The phrase will be read as the second. Disambiguate in the paper or the testbed appears broken.
- Vector AR(1) with multivariate- t innovations is Markov, hence Level 0. It can demonstrate scaling, never a gap.

Models that do produce a gap are those whose nonlinearity sits *above* the library’s granularity: banded-precision elliptical (Level 1), delay- k threshold AR with k beyond the window (Level 2), latent blocks or replicates (Level 2), and dynamic latents (Level 3).

2.4 (Q1) The elliptical theorem: nonlinear beats linear

Theorem 8 (Elliptical families with banded inverse scatter; TO VERIFY). *Let $Y \sim E_n\{0, \Sigma(\gamma), g_\nu\}$ with density $c(\theta)|\Sigma|^{-1/2}g_\nu\{y^\top \Sigma(\gamma)^{-1}y\}$, $\theta = (\gamma, \nu)$, and suppose $\Sigma(\gamma)^{-1}$ is b -banded for all γ . Then*

$$T = \left(\sum_t y_t y_{t+\ell} \right)_{0 \leq \ell \leq b}$$

is sufficient for θ and is a sum of pairwise functionals at lags $\leq b$, so (3) holds at pairwise granularity. If in addition the lag- ℓ bivariate elliptical sub-score is injective in $(A, C) = (x^2 + y^2, xy)$ modulo the sign symmetry respected by T (condition (4)), then:

- (i) *nCS on exact pairwise sub-scores up to lag b recovers g_0 exactly, including the generator parameter ν ;*
- (ii) *$G_\star^{\text{lin}} = I(\theta_0)$ if and only if the generator g_ν is Gaussian;*
- (iii) *in the scatter directions, $I_n(\gamma) = \Theta(n)$ and the deficit $e_\gamma^\top \{I - G_\star^{\text{lin}}\} e_\gamma$ is a fixed positive fraction of it, hence $\Theta(n)$: order-one per observation, permanently.*

Proof strategy. (i) is Theorem 1 with ψ the pairwise cross-products. (ii): g_0 is linear in T iff $\nabla_\theta \log g_\nu$ is affine, i.e. iff $g_\nu(s) \propto e^{-s/2}$. Gaussianity is the fixed point because only then is the

tilt affine (Corollary 5). (iii) follows from Proposition 4: the residual R has variance a fixed multiple of n in the scatter block under stationarity. *To verify:* the injectivity in (4) is checked numerically (zero collisions among 4,000 sampled bivariate- t sub-score vectors) but not proved; the sign symmetry must be stated in the theorem, not buried in the proof. $\square$

Remark 9 (Two boundaries, and a correction to an earlier draft). (a) *Banded inverse, not banded covariance.* An elliptical MA model has dense precision, so no finite-band T is sufficient; pairwise nCS then approximates with geometrically decaying truncation error in the bandwidth (a Level-2 $\rightarrow$ 3 transition).

(b) *The degenerate limit is Gaussianity, not decorrelation.* At γ giving $\Sigma = I$ the density is $c g_\nu(\sum_t y_t^2)$, which is *not* a product unless g_ν is Gaussian: the coordinates remain coupled through the common scale mixture (Theorem 10(a)). An earlier draft of this note asserted that $\rho \rightarrow 0$ closes the gap by factorization; that is true for an innovation/Markov model and false here.

(c) *The tail direction saturates.* For a *global* elliptical model there is one latent scale, so $I_n(\nu) = O(1)$ and no $\Theta(n)$ claim is available in the ν direction — see Proposition 12. The honest $\Theta(n)$ statement is (iii), in the scatter directions. An earlier draft stated the theorem in the ν direction; that was the one coordinate where it cannot hold.

(d) *Extension.* The corollary carries verbatim to elliptical copulas with parametric margins (banded-precision Gaussian or t copulas: the log-density is the elliptical quadratic in transformed scores plus separable marginal terms) — an untested prediction worth one confirmatory experiment.

2.5 (Q2) The latent dichotomy: when pre-aggregation nonlinearity is necessary

Theorem 10 (Conditional exponential families; latent geometry decides the level; PROVED). *Let $Y_i \mid Z$ be conditionally independent with $\log f(y_i \mid z) = \eta(z_{(i)})^\top T_i(y_i) + h_i(y_i) - A_i\{\eta(z_{(i)})\}$.*

- (a) Global latent ($z_{(i)} \equiv Z$): *the marginal likelihood depends on the data only through $\sum_i T_i(Y_i)$. Level 1: **pointwise nCS is exact**, with ϕ the inverse-saturation map $u_i \mapsto T_i$ and q the log-mixture gradient.*
- (b) Independent block latents ($z_{(i)} = Z_{b(i)}$, Z_b i.i.d., blocks in $\mathcal{U}$): *the likelihood factorizes over blocks and $g_0 = \sum_b g(T_b)$ with $T_b = \sum_{i \in b} T_i$. Level 2: **blockwise nCS is exact**; pointwise nCS is not, because g is nonlinear in the block aggregate.*
- (c) Dynamically coupled latents (Z a Markov chain): *the marginal likelihood is the matrix product $\pi_0^\top D_1 P D_2 \cdots P D_n \mathbf{1}$. Non-commutativity destroys additive separability, so by Theorem 2 **no finite separable sufficient statistic exists over any fixed-width covering**. What survives is an approximation law: smoothing distributions forget geometrically (Dobrushin/Birkhoff contraction), giving*

$$1 - \text{RE}\{\text{windowed nCS, width } w\} \lesssim C\rho^{2w}, \quad \rho \uparrow 1 \text{ as persistence } \uparrow.$$

Remark 11 (Hoeffding reading of (c)). The deficit of width- w nCS equals the energy of the score in interaction components of order $> w$. Dynamic latents are exactly the models whose score has geometrically decaying but never-vanishing high-order components — which is why pointwise ϕ recovers essentially nothing in sequential models and why the required width scales like the mixing time.

Proposition 12 (Manifold collapse: where Method C suffices; PROVED). *Under Theorem 10(a) with $n \rightarrow \infty$: conditionally on $Z = z$ the batch sums obey an LLN, $CS/n \rightarrow \mu(z)$ a.s. If $z \mapsto \mu(z)$ is injective then CS asymptotically identifies Z , all global aggregates collapse onto the manifold $\{\mu(z)\}$, and the score — itself asymptotically a function of Z — becomes a function of CS:*

$$\text{RE}_C \rightarrow 1.$$

Consequently a single global latent **cannot** demonstrate (Q2): outer nonlinearity alone is asymptotically sufficient, and the feature networks $a_\phi, b_{\phi,k}$ are redundant there. Moreover the information about a parameter of the latent’s own distribution saturates: $I_n(\nu) = O(1)$, since the data identify one draw of Z .

Remark 13 (Concede this explicitly). Stating Proposition 12 strengthens the paper. It also explains the observed pattern that Method C nearly closes the gap on the multivariate- t and global mixture models ($\text{RE } 0.76 \rightarrow 0.97$ and $0.91 \rightarrow 0.98$ from A to C), with C’s edge over A growing in n as the manifold sharpens.

Theorem 14 (Blockwise gap: CLT washing; TO VERIFY). *Assume Theorem 10(b) with m i.i.d. blocks of fixed size J , $n = mJ$. Decompose the per-block score by L^2 projection onto the block statistics the global sums can track, $g(T_b) = a + B^\top T_b + e(T_b)$, and assume $(CS, \sum_b e(T_b))$ obeys a joint CLT. Then $\text{Cov}\{CS, \sum_b e(T_b)\} = 0$ by construction of the projection, asymptotic joint normality upgrades this to asymptotic independence, and*

$$\text{RE}_C \rightarrow \text{RE}_A \rightarrow 1 - \frac{\text{Var}\{e(T_b)\}}{\text{Var}\{g(T_b)\}} \quad \textbf{independent of } m,$$

while blockwise nCS attains $\text{RE} = 1$. Hence the information deficit of every linear and every outer-nonlinear aggregation is

$$I(\theta_0) - G_\star^{\text{lin}} = m \cdot \text{Var}\{e(T_b)\} \cdot \{1 + o(1)\} = \Theta(m) = \Theta(n),$$

order-one per block, permanently, and removed entirely by pre-aggregation nonlinearity at block granularity.

Remark 15 (Why this, and not the elliptical model, is the load-bearing case). Theorem 14 is the only statement in Part I that answers (Q2), and it is also the only one that survives Proposition 12. Its structure is favourable in three ways verified numerically: (i) $C \approx A$ to ≤ 0.01 in every block-latent experiment; (ii) the deficit is stable in m at fixed block size ($1 - \text{RE}_A \approx 0.067$ at $m=40$, 0.074 at $m=160$) while C’s edge over A *shrinks* ($0.005 \rightarrow 0.001$), exactly as CLT washing predicts; (iii) the deficit grows with block internal dimension ($J=4$: 0.026 ; $J=16$: 0.11 ; $d=5$: 0.06 ; $d=15$: 0.20), because larger blocks sharpen the per-block posterior of Z_b and increase the curvature of g relative to its linear part. *Profitable regime*: fixed moderate block size with $m \rightarrow \infty$. Fixed m with large J does not produce a growing gap. *To verify*: the joint CLT is assumed, not proved; the conclusion is asymptotic, so finite- m recoverability of the residual by Method C should be acknowledged.

Proposition 16 (A-priori screening diagnostic; PROVED, GIVEN THM 14). *The per-block nonlinearity ratio $\text{Var}\{e(T_b)\}/\text{Var}\{g(T_b)\}$ is estimable from pilot simulations at the block level, before any training, and by Theorem 14 it is the asymptotic gain of blockwise nCS over Methods A and C. For sequential models the Hoeffding decomposition extends this to a window-order spectrum. This is the cheapest available answer to “is the gap worth chasing on this model?”*

2.6 Motivating examples

The following are ordered as they should appear in the paper: two that establish (Q1), one that establishes (Q2) and carries the headline, one negative control, and two that delimit scope.

Example 1 (Elliptical AR-scatter field — the (Q1) workhorse). $Y \sim t_\nu\{0, \Sigma(\rho)\}$ with $\Sigma(\rho)^{-1}$ tridiagonal (AR(1) scatter). Level 1. Closed-form g_0 , so the gap is measurable against ground truth. Theorem 8: nCS on lag- ≤ 1 pairwise sub-scores is exact; linear CS is not, with deficit a fixed fraction of $\Theta(n)$ information in the ρ direction. Measured $\text{RE}_A \approx 0.76\text{--}0.85$ at $\nu = 3$ versus $0.87\text{--}0.93$ at $\nu = 10$, both $\rightarrow 1$ under the oracle — the ν -dependence is the distance from the Gaussian fixed point. **Must be labelled unambiguously as the elliptical model**, never as “Student- t AR(1)” (Remark 7). Report the ρ direction, not ν (Remark 9(c)).

Example 2 (Global regime mixture — exact-inversion illustration). $Y_i | Z \sim (1 - Z)f_0 + Zf_1$ with a single global regime indicator. Level 1, Theorem 10(a). The recovery map is available in closed form: $\phi(s) = \log\{1 + (1 - \pi)s\} / \{1 - \pi s\}$ recovers $\log r_i$ exactly, giving oracle $\text{RE} = 1.000$. Its value is pedagogical — it exhibits ϕ as the *inverse of the sub-score’s saturation* (Principle 2) — and it is an honest illustration of Proposition 12: Method C reaches 0.98, so this example must **not** be used to argue for pre-aggregation nonlinearity.

Example 3 (Independent block latents — the headline (Q2) example). m i.i.d. blocks of size J , each with its own latent: blockwise- t (per-block scale mixture), common-shock ($Y_{b,i} = \lambda_b + \varepsilon_{b,i}$, d -dimensional), or max-stable replicates. Level 2, Theorem 14. This is the load-bearing example: information $\Theta(m)$ with no saturation, deficit stable in m and tunable through J or d , exact recovery by hierarchical per-block pooling and *not* by global pooling, and $\text{RE}_C \approx \text{RE}_A$ so the comparison isolates pre-aggregation nonlinearity. Measured: block- t $0.89 \rightarrow 0.99$; common shock $d=15$ $0.80 \rightarrow 0.99$; max-stable $0.78 \rightarrow 1.000$ (identical to four decimals between A and C).

Example 4 (Gaussian banded-precision field — negative control). Same design as Example 1 with a Gaussian generator. By Theorem 8(ii) the gap is *exactly* zero, and by Proposition 6 any first-order Markov model with a pairwise+marginal library is likewise exactly zero. A method that beats the linear ceiling here is reporting noise. Two independent negative controls — generator Gaussianity and Level-0 separability — are stronger than one.

Example 5 (Delay- k threshold AR — granularity, not nonlinearity). $Y_t = \alpha Y_{t-1} + \beta Y_{t-1} \mathbf{1}\{Y_{t-k} > c\} + \varepsilon_t$ with delay k exceeding the library’s window. Level 2. With triple margins covering the delay, linear CS is exact ($R^2 = 1.000000$, Proposition 6); with only pairwise margins, pointwise ϕ gains essentially nothing ($0.892 \rightarrow 0.900$) while windowed ϕ at matched granularity gains $+0.09$ ($\rightarrow 0.98$ against a triple-margin ceiling of 0.99). The cleanest available demonstration of Principle 1: the batching structure, not the flexibility of ϕ , is what buys recovery.

Example 6 (HMM / stochastic volatility — Level 3 and the window law). Two-state mean- or variance-switching HMM, or stochastic volatility with quadrature-exact bivariate sub-scores. Level 3, Theorem 10(c): no finite exact recoverer, geometric window law $1 - \text{RE} \lesssim C\rho^{2w}$. Measured on the persistence parameter: 0.18 (pointwise) $\rightarrow$ 0.67 (windows, $b=2$) $\rightarrow$ 0.80 (with lags), pointwise ϕ contributing $+0.00$. This example earns its place by supplying a *rate* rather than a dichotomy, and by motivating recurrent or attention aggregation as the Level-3 completion. Predictions worth one confirmatory run: Hawkes processes are Level 3 with window scale $(1 - \alpha/\kappa)^{-1}$, so near-criticality plays the role of $\rho \uparrow 1$.

2.7 Two architecture principles with theoretical warrant

Principle 1 (Granularity matching). ϕ contributes nothing below the granularity of the configuration unit carrying the nonlinearity — the latent’s support, the interaction clique, the mixing window. Formally, $\mathcal{N}(\mathcal{U})$ contains only $\sigma\{\sum_i \phi(u_i)\}$ -measurable statistics; if the separable sufficient T requires summands on cliques outside $\mathcal{U}$, no expressiveness of ϕ compensates. Verified three times independently: pointwise ϕ gained ≈ 0 over Method C in the mean-switching HMM ($0.243 \rightarrow 0.232$), the variance-switching HMM ($0.243 \rightarrow 0.232$), and the threshold AR ($0.892 \rightarrow 0.900$), while window/block ϕ at matched granularity gained $+0.42$, $+0.42$, $+0.09$. *Consequence: the batch structure of $\text{CS}_{2,k}$ is not an implementation detail but the central structural hyperparameter, and should be exposed and tuned (pointwise $\rightarrow$ sliding windows $\rightarrow$ blocks $\rightarrow$ recurrent).*

Principle 2 (Saturation inversion). Exact sub-scores of mixtures and heavy-tailed families are *bounded*, *redescending* transforms of unbounded local evidence: the $\log \sigma$ -component of the t sub-score saturates at $\nu + \dim$, mixture scores at $1/\pi$. The sufficient statistics ($\sum r_i$, $\sum \log r_i$, Q , $(\sum y)^2$) are unbounded. Linear sums of bounded images cannot represent unbounded statistics — this is the *mechanism* of linear failure — and the remedy ϕ must invert the saturation. *Consequence: trained ϕ networks need unbounded or exponential-type output links.* Bounded dictionaries systematically under-recover tails (max-stable: dictionary 0.85 vs structured features 1.000; global mixture: dictionary $\approx$ Method C vs inversion oracle 1.000).

Remark 17 (When condition (4) fails outright). Local score-sufficiency can fail for reasons no ϕ repairs: the unit-Fréchet marginal sub-score is *identically zero* for the dependence parameter, so the library must be enlarged (pairs, replicates), not the nonlinearity. Conversely (4) can often be repaired without new densities by evaluating sub-scores at several pilots $\beta_1, \dots, \beta_K$: the score field at multiple points determines the local likelihood shape and typically restores injectivity (e.g. when a single-pilot sub-score is even and an odd statistic is needed). This is the same multi-pilot panel that appears in Part II as an amortization option, now with a recovery-theoretic rationale rather than a brute-force one.

Remark 18 (Scope discipline). All statements in Part I are population-level information-geometry claims: they bound what architectures *can represent*, not what optimizers find. Reported training slack (of order ten points in the original Method-C experiment) belongs in the empirical section, not here. Claims should be flagged as holding (a) at population level, (b) under finite capacity, or (c) under a specific model.

3 Part II: the auxiliary-parameter problem

The surrogate score is a two-argument object $S_{\psi, \phi}(x, \beta)$. Mode A uses (Section 5) query it on the diagonal $\beta = \theta$; Mode B needs a θ -free statistic, which forces $\beta = \hat{\beta}(X)$ to be data-dependent. This section characterizes completely what can and cannot be done about the resulting pilot dependence.

3.1 The translation identity

Lemma 19 (Translation identity; PROVED). *Under Assumptions 1–2,*

$$\partial_{\beta} \mu(\theta, \beta) \big|_{\beta=\theta} = -H(\theta) = -\partial_{\theta} \mu(\theta, \beta) \big|_{\beta=\theta}.$$

Consequently, locally $\mu(\theta, \beta) = H(\theta - \beta) + o(\|\theta - \beta\|)$: the mean of the estimating function depends on θ and β only through their difference.

Proof. Differentiate $\mu(\theta, \theta) = 0$ (Assumption 2) with respect to θ by the chain rule: $\partial_\theta \mu + \partial_\beta \mu = 0$ on the diagonal. Rearranging gives the claim; a first-order Taylor expansion about $\beta = \theta$ gives the local form. $\square$

Lemma 19 is the engine of everything below. Note it is not a property of the architecture: the true score satisfies it ($\partial_\theta \mathbb{E}_\theta g_0 = I = -\mathbb{E} \partial_\beta g_0$, the second Bartlett identity in disguise), and any Fisher-consistent surrogate inherits it. Sensitivity to the data-generating parameter and sensitivity to the evaluation point are the same matrix with opposite sign.

Lemma 20 (Local linearization; PROVED). *Under Assumption 1, uniformly for β in an $O(n^{-1/2})$ -neighbourhood of θ_0 ,*

$$m(X, \beta) = H(\hat{\theta}_1 - \beta) + o_p(n^{1/2}), \quad (6)$$

where $\hat{\theta}_1 = \beta + H^{-1}m(X, \beta)$ is the one-step estimator from β and the remainder is uniform over the neighbourhood.

Proof. Immediate from (1) as an identity, with the stochastic remainder controlled by expanding $m(X, \beta)$ around θ_0 and applying Lemma 19: the $O_p(\|\beta - \theta_0\|)$ drift is exactly $-H(\beta - \theta_0)$, so $m(X, \beta) = m(X, \theta_0) - H(\beta - \theta_0) + o_p(n^{1/2})$, and the definition of $\hat{\theta}_1$ absorbs both terms. Uniformity follows from stochastic equicontinuity. $\square$

Equation (6) is worth reading slowly. The score value at any evaluation point is, to first order, the discrepancy between the one-step estimator and that point, scaled by H . The taxonomy of algorithms in Section 5 is (6) evaluated at three choices of β .

3.2 Impossibility and characterization

Call a statistic F *first-order pilot-invariant* if its total derivative along a perturbation of the evaluation point vanishes at leading order.

Theorem 21 (Impossibility and characterization; PROVED). *Let H be nonsingular and let F be C^1 .*

- (a) (Impossibility.) *Suppose F depends on the pair $(\hat{\beta}, m(X, \hat{\beta}))$ only through the second argument, $F = F(m)$. If F is first-order pilot-invariant then $D_F H = 0$, hence $D_F = 0$, and F is first-order ancillary: its mean does not respond to θ at leading order.*
- (b) (Characterization.) *Suppose $F = F(\hat{\beta}, m)$. Then F is first-order pilot-invariant if and only if it solves*

$$F_\beta = F_m H,$$

whose general solution is

$$F(\hat{\beta}, m) = \Phi(\hat{\beta} + H^{-1}m) = \Phi(\hat{\theta}_1)$$

for an arbitrary C^1 function Φ .

Proof. Perturb the evaluation point $\hat{\beta} \mapsto \hat{\beta} + d\beta$. By Lemma 19 the mean of m responds as $dm = -H d\beta$, while the response to a perturbation of the data-generating parameter is $+H d\theta$.

(a) With $F = F(m)$, $dF = F_m dm = -F_m H d\beta$. Pilot-invariance requires $F_m H = 0$ for all $d\beta$, so $F_m = 0$ by nonsingularity of H . But the response of F to θ is $F_m H d\theta$, which then also vanishes. Hence F carries no first-order information about θ .

(b) With $F = F(\hat{\beta}, m)$, $dF = F_\beta d\beta + F_m dm = (F_\beta - F_m H) d\beta$. Invariance is therefore $F_\beta = F_m H$, a linear first-order PDE. Its characteristic curves are $\{(\beta, m) : \beta + H^{-1}m = \text{const}\}$, since along $\dot{\beta} = v$, $\dot{m} = -Hv$ the combination $\beta + H^{-1}m$ is constant. The general solution of a linear homogeneous transport equation is an arbitrary function constant on characteristics, giving $F = \Phi(\beta + H^{-1}m) = \Phi(\hat{\theta}_1)$. $\square$

Remark 22 (Interpretation). Part (a) says: *pilot-invariance and θ -informativeness are the same derivative, with opposite sign.* Trying to launder the pilot error out of the score value alone necessarily launders out the signal. In particular there is no “debiased score” summary — a natural reviewer suggestion — and no rescue from over-identification ($q > p$): directions in the left null space of H are pilot-insensitive precisely because they are θ -insensitive.

Part (b) says: the one-step estimator is *not one option among several*. It is the unique first-order pilot-invariant coordinate on the pair, up to bijection. Any procedure that removes pilot dependence while retaining information — including “improve $\hat{\beta}$ to $\tilde{\beta}$ using the surface, then evaluate $S_{\psi,\phi}(X, \tilde{\beta})$ ” — reconstructs $\hat{\theta}_1$ or fails.

Corollary 23 (Degeneracy at the corrected point; PROVED). $m(X, \hat{\theta}_1) = o_p(n^{1/2})$; that is, the score evaluated at the one-step estimator is first-order pilot-invariant but carries only second-order content. Iterating, $m(X, \hat{\theta}_k)$ is $o_p(n^{1/2})$ for every $k \geq 1$, and at the exact root $m(X, \tilde{\theta}) \equiv 0$ identically. Thus $S_{\psi,\phi}(X, \tilde{\theta}) = 0$ is the limiting case ($k \rightarrow \infty$) of the same phenomenon.

Proof. Set $\beta = \hat{\theta}_1$ in (6): the leading term is $H(\hat{\theta}_1 - \hat{\theta}_1) = 0$. The root is the fixed point of the iteration, at which the estimating equation holds by definition. $\square$

Corollary 24 (Information conservation; PROVED). Along the Newton trajectory $\beta \mapsto \hat{\theta}_1(\beta)$, the first-order information carried by the score value decreases monotonically to zero, while the information in the pair $(\beta, m(X, \beta))$ is conserved: for every β in the tube,

$$\sigma(\hat{\theta}_1) \subseteq \sigma(\beta, m(X, \beta)),$$

and the two σ -fields carry the same first-order information about θ . Information is not lost as $\beta \rightarrow \tilde{\theta}$; it migrates from the score value into the evaluation point.

Remark 25 (The summary must not be the score value alone). Corollaries 23–24 jointly foreclose using $S_{\psi,\phi}(X, \hat{\beta}(X))$ without $\hat{\beta}$ as an amortized summary: the better the pilot, the less informative the summary, degenerating to the zero statistic at the root. This does *not* apply at a fixed evaluation point β_{obs} (see Proposition 34).

3.3 Consequences for summary construction

Proposition 26 (Sufficiency ordering; PROVED). $\sigma(\hat{\theta}_1) \subseteq \sigma(\hat{\beta}, m(X, \hat{\beta}))$, hence for any prior,

$$\text{the posterior } p(\theta \mid \hat{\beta}, m(X, \hat{\beta})) \text{ is at least as sharp as } p(\theta \mid \hat{\theta}_1),$$

with first-order equality: the extra coordinate is asymptotically ancillary. Both dominate $p(\theta \mid m(X, \hat{\beta}))$ strictly, by Corollary 23.

Proof. $\hat{\theta}_1$ is a fixed measurable (indeed affine) function of the pair, so its generated σ -field is coarser; conditioning on a finer σ -field cannot lose information. First-order equality is Theorem 21(b) applied with $\Phi = \text{id}$: the residual coordinate is invariant-complementary, and by Lemma 20 it equals $m(X, \hat{\theta}_1) = o_p(n^{1/2})$. $\square$

Remark 27 (Which to feed the density estimator). Proposition 26 favours the pair on information grounds; two countervailing considerations favour $\hat{\theta}_1$. (i) The map $(\hat{\beta}, m) \mapsto \hat{\theta}_1$ is a *first-order linearization*, exact only to the order of (6); feeding the pair lets the density estimator learn the true nonlinear invariant combination, which matters precisely in the wide-tube / bad-pilot regime. (ii) Conversely, feeding $\hat{\theta}_1$ injects a known combination as inductive bias, saving the estimator from rediscovering it — an advantage only at small simulation budgets. (iii) The analytic sandwich and the efficiency statement are properties of $\hat{\theta}_1$; the UQ claim must be anchored to it. Recommendation: pair as the method, $\hat{\theta}_1$ as theory anchor and small-budget ablation.

Remark 28 (Centering and NPE-invariance). Since $\mathbb{E}_\beta S_{\psi, \phi}(X, \beta)$ is a known function of β , replacing $S_{\psi, \phi}$ by m is a bijection of the summary vector *when $\hat{\beta}$ is included*. Hence the NPE posterior is invariant to centering. Centering is nonetheless required for: $\hat{\theta}_1$, H, V , the sandwich, and every asymptotic statement above. FSM training (Algorithm 1) delivers approximate centering automatically, since the population optimum is the true score; the residual $\|\widehat{\mathbb{E}_\beta S_{\psi, \phi}}\|$ is therefore a free diagnostic of approximation error.

4 Part III: validity, efficiency, and pilot conditions

Proposition 29 (Validity is unconditional; PROVED). *Let $\hat{\beta} : \mathcal{X} \rightarrow \mathbb{R}^p$ be any measurable map — consistent, inconsistent, or arbitrary. Then $T(X) = (\hat{\beta}(X), S_{\psi, \phi}(X, \hat{\beta}(X)))$ is a statistic of X alone, and $p(\theta \mid T(X_{\text{obs}}))$ is an exact posterior given T . Pilot error introduces no bias into SBI inference.*

Proof. Composition of measurable maps. The posterior given any statistic is well defined and exact for that statistic; nothing about its construction need reference θ . $\square$

Remark 30 (The deployability pillar, stated correctly). Proposition 29 is the precise content of “pilot robustness”. SBI conditioning converts pilot error from a *validity* problem into an *efficiency* problem. This is a categorical difference from the classical one-step regime, where a bad pilot yields a wrong estimator and wrong sandwich intervals. The claim to make is therefore not “we mitigate bad pilots” but “conditioning makes pilot error an efficiency question, and the tube plus the invariant coordinate recover the efficiency.” The narrower phrase “the main obstacle to *amortizing* score-based summaries” should be preferred to “...to using score-based summaries”, since at a fixed evaluation point the obstacle largely disappears (Proposition 34).

Condition 1 (Tube coverage). Let Π_{train} be the law of (X, β) generated by Algorithm 1 and Π_{deploy} the law of $(X, \hat{\beta}(X))$ under $\theta \sim \pi$. Then $\Pi_{\text{deploy}} \ll \Pi_{\text{train}}$ with bounded Radon–Nikodym derivative on a neighbourhood of the diagonal containing the pilot’s support.

Proposition 31 (Tube approximation; TO VERIFY). *Suppose the trained surface satisfies $\|S_{\psi, \phi} - g_0\|_{L^2(\Pi_{\text{train}})} \leq \epsilon$ and Condition 1 holds with RN-derivative bounded by C . Then the pair summary attains Godambe information at least $G - O(\sqrt{C}\epsilon)$ (approximation error can only lose information, never add it), and the one-step estimator satisfies*

$$\hat{\theta}_1 - \theta_0 = I(\theta_0)^{-1} \frac{1}{n} g_0(X; \theta_0) + O_p(\|\hat{\beta} - \theta_0\|^2) + O_p(\epsilon) + o_p(n^{-1/2}).$$

Remark 32. Condition 1 has no classical analogue: the true score is globally exact, so no train/deploy distributional condition is ever needed. It is the price of a *learned* surface, and it is checkable by simulation, unlike “the network is good”. It also dictates the design rule: **tube width** $\propto$ **pilot sandwich standard error**, with training density concentrated near the diagonal where later Newton iterates live. Too narrow reproduces the diagonal-only extrapolation failure; too wide spends capacity on the fixed- δ regime where the deterministic drift $-n\nabla_\beta \text{KL}$ dominates the $O_p(\sqrt{n})$ stochastic signal.

Proposition 33 (One-step and k -step efficiency; PROVED). *Assume Assumption 1 and $\epsilon = 0$. If $\|\hat{\beta} - \theta_0\| = o_p(n^{-1/4})$ then $\hat{\theta}_1$ is first-order efficient for the estimating function:*

$$\sqrt{n}(\hat{\theta}_1 - \theta_0) \rightsquigarrow N(0, G^{-1}), \quad G = H^\top V^{-1} H,$$

and $\hat{\theta}_1$ is asymptotically equivalent to the root $\tilde{\theta}$; if $S_{\psi, \phi} = g_0$ then $G = I(\theta_0)$ and $\hat{\theta}_1$ is asymptotically equivalent to the MLE. If instead $\|\hat{\beta} - \theta_0\| = O_p(n^{-\gamma})$ with $\gamma \in (0, 1/4]$, then $k = \lceil \log_2\{1/(2\gamma)\} \rceil$ Newton steps suffice for the same conclusion.

Proof sketch. Expand $m(X, \hat{\beta})$ around θ_0 using Lemma 19: the linear term $-H(\hat{\beta} - \theta_0)$ cancels the $\hat{\beta}$ outside (1), leaving $\hat{\theta}_1 - \theta_0 = H^{-1}m(X, \theta_0) + O_p(\|\hat{\beta} - \theta_0\|^2)$. The quadratic remainder is $o_p(n^{-1/2})$ iff $\|\hat{\beta} - \theta_0\| = o_p(n^{-1/4})$. Newton error squares per iteration, giving the stated k . Asymptotic normality with covariance $H^{-1}VH^{-\top}$ and $G = H^\top V^{-1}H$ is the standard sandwich. $\square$

Proposition 34 (Fixed evaluation point; PROVED). *Let $\beta_{\text{obs}} = \hat{\beta}(Y_{\text{obs}})$ be computed once and held fixed across pseudo-datasets. Then:*

- (i) $x \mapsto m(x, \beta_{\text{obs}})$ is a deterministic statistic and, exactly — by definition of the one-step map $\hat{\theta}_1 = \beta_{\text{obs}} + \hat{H}^{-1}m(x, \beta_{\text{obs}})$, with β_{obs} and $\hat{H}$ constants across pseudo-datasets, no linearization needed — an affine bijection of $\hat{\theta}_1$; the degeneracy of Corollary 23 does not activate.
- (ii) Consequently ABC with an affine-invariant discrepancy gives identical (not merely asymptotically equivalent) output whether the summary is $m(\cdot, \beta_{\text{obs}})$ or $\hat{\theta}_1$.
- (iii) Consistency of $\hat{\beta}$ at any rate suffices for first-order efficiency: the information loss is $O(\|\beta_{\text{obs}} - \theta_0\|)$ and no expansion in the pilot error is inverted.

Remark 35 (Why the two regimes differ). Proposition 34(iii) versus Proposition 33 answers a natural question: why does ABC need only consistency while the amortized one-step needs a rate? Because ABC at fixed β never *inverts* the pilot error; the Newton map does, and inversion is what imposes $o_p(n^{-1/4})$.

4.1 Tiered pilot conditions

Remark 36 (Pilot design under heavy tails). For Student- t with $\nu \in (3, 4]$ the natural moment pilot for ν (sample kurtosis) requires a finite fourth moment and is inconsistent or arbitrarily slow. Use rank- or quantile-based pilots: Kendall’s τ / Gaussian-rank correlation for ρ , a quantile-ratio or tail-index estimator for ν . Both are closed-form and $\sqrt{n}$ -consistent without moment conditions. This detail will be checked by any referee who knows t -models.

Guarantee	Requires	Algorithm	Reference
Posterior validity	measurability of $\hat{\beta}$ only	6	Prop. 29
Surface accuracy at deployment	tube coverage (C1)	1, 6	Prop. 31
Non-degenerate information	identifiability of θ from $\mu(\cdot, \beta^*)$	6	Cor. 23
Efficiency, ABC at fixed β	consistency, any rate	8	Prop. 34
Efficiency + analytic sandwich	$o_p(n^{-1/4})$, H nonsingular, correct basin	7, 4	Prop. 33

Table 2: What each guarantee costs. Note that pilot *rate* enters only the last row; validity is free, and the fixed- β regime is rate-insensitive.

5 Part IV: algorithms

The organizing axis is the evaluation point β . Mode A queries the surface on the diagonal $\beta = \theta$ (no pilot needed). Mode B needs a θ -free map, obtained by collapsing β : freeze it, retain both coordinates, or Newton-correct it into the invariant coordinate.

5.1 Training

Algorithm 1: Tube training of the score surface (recommended)

Input: prior/grid for tube centers; tube width $\tau \propto$ pilot sandwich SE; batch B

```

1 repeat
2   sample center  $\beta$ ;
3   sample  $\theta \sim N(\beta, \tau^2 I)$  ;                               //  $\theta$  downstream of  $\beta$  --- critical
4   simulate  $X \sim p(\cdot | \theta)$ ;
5   target  $Y \leftarrow (\theta - \beta)/\tau^2$ ;   predict  $\hat{S} \leftarrow S_\psi\{T_\phi(X, \beta)\}$ ;
6   gradient step on  $(\psi, \phi)$  to reduce  $\|\hat{S} - Y\|^2$ ;

```

```

7 until converged;

```

Output: $S_{\psi, \phi}(x, \beta) \approx \nabla_\beta \log p(x | \beta)$, accurate on the tube

Why β must be an ancestor of X . The FSM identity $\mathbb{E}[(\theta - \beta)/\tau^2 | X] \approx \nabla_\beta \log p(X | \beta)$ requires the generative order $\beta \rightarrow \theta \rightarrow X$. If instead $\beta = \hat{\beta}(X)$ is computed *from* X (a descendant), the population minimizer becomes $\{\mathbb{E}[\theta | T_\phi(X)] - \hat{\beta}(X)\}/\tau^2$, i.e. the *posterior-mean correction*, not the score. That object is Algorithm 9, a different (and weaker) method. Evaluating a tube-trained surface at $\hat{\beta}(X)$ after training is fine; *training against a target defined through $\hat{\beta}(X)$* silently changes the estimand.

Two β -sourcing variants inside the loop: *surrogate* $\beta = \theta + \varepsilon$, $\varepsilon \sim N(0, \hat{\Sigma}_{\text{pilot}}/n)$ (training-only shortcut; second-order train/deploy mismatch since it makes $\beta \perp\!\!\!\perp X | \theta$ and exactly Gaussian); or *real pilot* $\beta = \hat{\beta}(X)$ (exact pilot geometry, costs one pilot solve per simulation). The surrogate is admissible *only* inside Algorithm 1; it is inadmissible in Mode B training (Section 5.4), where the summary map must be identical at NPE-training and observation time.

Algorithm 2: Godambe information maximization (alternative / fine-tuning objective)

Input: training points θ ; batch B ; ridge λ ; step δ

```

1 repeat
2   simulate  $X^{(1:B)} \sim p(\cdot \mid \theta)$ ;  $S_b \leftarrow S_{\psi,\phi}(X^{(b)}, \theta)$ ;
3    $\hat{V} \leftarrow \widehat{\text{Cov}}(S_b) + \lambda I$ ;
4   for  $k = 1, \dots, p$  do
5      $\hat{H}_{:,k} \leftarrow \{\hat{\mu}_S(\theta + \delta e_k) - \hat{\mu}_S(\theta - \delta e_k)\} / (2\delta)$ ;           // or autodiff / CRN
6   end
7   maximize  $\mathcal{L} = 2 \log |\det \hat{H}| - \log \det \hat{V}$ ;                               // or variational
      maximize  $\max_M \text{tr}(2M^\top \hat{H} - M^\top \hat{V} M)$ 
8 until converged;
```

Both objectives target the same population optimum (the Fisher score, up to nonsingular linear transformation); Algorithm 1 is cheaper and inverse-free, Algorithm 2 targets efficiency directly. Use 2 as a gating check: Monte-Carlo $I(\theta_0) - G_\star^{\text{lin}}$ before investing in the full pipeline.

5.2 Shared components

Algorithm 3: Li–Fearnhead regression head (inverse-free Newton coefficient)

Input: tube simulations $\{(\theta^{(b)}, \beta^{(b)}, X^{(b)})\}$ from Algorithm 1

```
1  $\hat{B} \leftarrow \arg \min_B \sum_b \|\theta^{(b)} - \beta^{(b)} - B S_{\psi,\phi}(X^{(b)}, \beta^{(b)})\|^2$ ;
```

Output: $\hat{B}$, estimating H^{-1} without forming or inverting $\hat{H}$

The population minimizer is the posterior-mean map, which by Lemma 19 coincides with H^{-1} to first order. This is the numerically preferred implementation: no finite differences, no δ , no matrix inverse, and it reuses simulations already generated. It is also, read differently, Fearnhead–Prangle semi-automatic regression applied to a learned structured summary.

5.3 Mode A: use the surface directly (no pilot)

Algorithm 4: Estimating-equation inference (k -step Fisher scoring)

```

1  $\theta^{(0)} \leftarrow \hat{\beta}(X_{\text{obs}})$ ;
2 for  $t = 0, \dots, k-1$  do
3    $m^{(t)} \leftarrow S_{\psi,\phi}(X_{\text{obs}}, \theta^{(t)}) - \mathbb{E}_{\theta^{(t)}} S_{\psi,\phi}$ ;           // centered; Remark 28
4    $\theta^{(t+1)} \leftarrow \theta^{(t)} + \hat{B} m^{(t)}$  (or  $\hat{H}^{-1} m^{(t)}$ );
5 end
Output:  $\hat{\theta} = \theta^{(k)}$ ; sandwich  $\hat{\Sigma} = \hat{H}^{-1} \hat{V} \hat{H}^{-\top}$  at  $\hat{\theta}$ 
```

$k = 1$ suffices under $o_p(n^{-1/4})$; iterate to convergence for the classical Z -estimator root (Proposition 33). Each step is a forward pass and a $p \times p$ solve — no new simulations.

Algorithm 5: Score-based sampler

```
1 Run Langevin / diffusion sampler with drift  $S_{\psi,\phi}(X_{\text{obs}}, \theta_{\text{current}})$ ;
```

The sampler queries the surface *along its own trajectory*, so $\beta = \theta$ throughout: the diagonal is exactly where Algorithm 1 trains, and no auxiliary-parameter problem arises. This is the cleanest use of the two-argument surface and the reason Part II is a Mode-B issue only.

5.4 Mode B: amortized SBI summaries

In every Mode-B algorithm the summary map must be *identical* at NPE-training and observation time; hence $\hat{\beta}$ is evaluated on every simulated dataset, and the surrogate $\beta = \theta + \varepsilon$ is *not* admissible here.

Algorithm 6: Pair summary $\rightarrow$ NPE (headline method)

```

1 summary( $X$ ):  $\beta \leftarrow \hat{\beta}(X)$ ; return  $(\beta, S_{\psi,\phi}(X, \beta)) \in \mathbb{R}^{2p}$ ;
2 simulate  $\theta^{(b)} \sim \pi$ ,  $X^{(b)} \sim p(\cdot \mid \theta^{(b)})$ ; run  $\hat{\beta}$  on each;
3 train  $q_\eta(\theta \mid \text{summary}(X))$  on  $\{(\theta^{(b)}, \text{summary}(X^{(b)}))\}$ ;
Output:  $q_\eta(\theta \mid \text{summary}(X_{\text{obs}}))$ 

```

Justification. Proposition 26 (weakly dominant, loses nothing); Proposition 29 (valid for any pilot); Theorem 21(b) (what the density estimator asymptotically learns is a function of $\hat{\theta}_1$ — so the black box is characterized, not merely trusted). The pair repairs the *structural insufficiency* of the score value alone under *any* random pilot, good or bad: (6) is one p -vector constraining two unknowns, and $\hat{\beta}$ closes the system. Pilot quality enters only via Condition 1 and NPE difficulty — never via validity. No tuning parameter k , no $\hat{H}$, no δ .

Algorithm 7: One-step summary $\rightarrow$ NPE (theory anchor / UQ / small budgets)

```

1 summary( $X$ ):  $\beta \leftarrow \hat{\beta}(X)$ ; return  $\hat{\theta}_1 = \beta + \hat{B} S_{\psi,\phi}(X, \beta) \in \mathbb{R}^p$ ;
2 train and deploy NPE as in Algorithm 6;
3 report  $\hat{\theta}_1$  as point estimate with sandwich  $\hat{\Sigma}$ ; check NPE spread against  $\hat{\Sigma}$ ;

```

Justification. Proposition 33 ($\hat{\theta}_1$ attains G ; Fearnhead–Prangle-optimal, since an efficient estimator is asymptotically sufficient and $\hat{\theta}_1 \simeq \mathbb{E}[\theta \mid T_\phi]$ by Li–Fearnhead). The sandwich describes *this* estimator, so conditioning on $\hat{\theta}_1$ keeps the analytic covariance and the posterior referring to the same object — the coherence the certification claim requires. NPE adds what the sandwich cannot: the finite-sample conditional law (skewness, tails, prior influence). Agreement between the two is the certificate.

Algorithm 8: Fixed- β_{obs} score summary $\rightarrow$ ABC (single dataset)

```

1  $\beta_{\text{obs}} \leftarrow \hat{\beta}(Y_{\text{obs}})$ ; // computed once
2 summary( $X$ )  $\leftarrow S_{\psi,\phi}(X, \beta_{\text{obs}})$ ; // same fixed point for all pseudo-data
3 run ABC comparing  $\text{summary}(Y_{\text{obs}})$  against  $\text{summary}(X^{(b)})$ ;

```

Justification. Proposition 34: the score value is an affine bijection of $\hat{\theta}_1$ here, so the degeneracy never activates and the score alone *is* a valid summary. Rate-insensitive to the pilot. Non-amortized: the per-dataset pilot cost is zero, but one NPE cannot serve future observations.

Algorithm 9: Baseline: Fearnhead–Prangle regression with composite-score features

```

1 simulate  $\theta^{(b)} \sim \pi$ ,  $X^{(b)} \sim p(\cdot \mid \theta^{(b)})$ ;
2 fit  $m_\eta: T_\phi(X, \hat{\beta}(X)) \mapsto \theta$ ; // targets  $\mathbb{E}[\theta \mid T_\phi(X)]$ , not the score
3 summary( $X$ )  $\leftarrow m_\eta(T_\phi(X, \hat{\beta}(X)))$ ; run NPE/ABC;

```

Role. Strongest Mode-B baseline, sharing the feature architecture, so the comparison isolates the value of the surface-plus-invariant-coordinate structure. It is simpler (no tube, no Newton, no $\hat{H}$) but forfeits: Mode A entirely (no two-argument surface), the k -step repair, prior-freeness (its target is defined under the training prior; the score surface is a frequentist object), the LAN inductive bias (it must model the posterior-mean map globally rather than the score locally), and any analytic covariance. Its UQ column is empty by construction.

6 Mapping claims to experiments

Each row is a proposition and the figure that must exist for it.

7 Open items

1. **Theorem 14** (the headline claim) needs the joint CLT for $(CS, \sum_b e(T_b))$ stated as a lemma rather than assumed, and the finite- m residual recoverability by Method C quantified. **Theorem 8** needs the injectivity in (4) proved rather than checked numerically, with the sign symmetry in the statement.
2. Confirm the Level-0 boundary empirically before fixing the testbeds (Remark 7): verify that AR(1) with i.i.d. t innovations and VAR(1) with multivariate- t innovations show *zero* gap under a pairwise+marginal library. A predicted zero that is observed is evidence the framework is right; an unexplained nonzero would indicate a library or implementation error.
3. **Proposition 31** needs the $O(\sqrt{C}\epsilon)$ constant made explicit, or restatement as a qualitative consistency claim.
4. Population equivalence of the Algorithm 1 and Algorithm 2 objectives at the optimum (worked out separately) should be included as a proposition; it licenses training with the cheaper FSM loss while claiming the Godambe conclusion.
5. Whether Theorem 21(b) extends beyond first order — i.e. whether an exactly pilot-invariant statistic exists — is open and probably negative.

Framing note. Lead with the dichotomy of Corollary 5: *the linear–nonlinear gap is exactly the curvature of the exponential tilt in its separable statistic*, so composite likelihood is lossless precisely at the exponential-family fixed point and nowhere else. Then the two claims, kept apart: Theorem 8 (nonlinear beats linear, elliptical families, iff-Gaussian boundary) and Theorem 14 (*pre-aggregation* nonlinearity beats outer nonlinearity, many independent latent units, with CLT washing showing outer nonlinearity provably cannot substitute). Concede Proposition 12 explicitly — a single global latent belongs to Method C — since doing so pre-empts the sharpest objection and costs nothing. Then: *exploiting the gap in amortized SBI requires resolving the auxiliary-parameter problem, which Theorem 21 settles completely*. Then the algorithms as consequences. The one-step machinery, LAN expansion, and Godambe training are classical recombinations — necessary, but not the novelty, and the abstract must not suggest otherwise.

References

- V. P. Godambe. An optimum property of regular maximum likelihood estimation. *Ann. Math. Statist.*, 1960.
- C. C. Heyde. *Quasi-Likelihood and Its Application*. Springer, 1997.
- W. Li and P. Fearnhead. On the asymptotic efficiency of approximate Bayesian computation estimators. *Biometrika*, 2018.

- P. Fearnhead and D. Prangle. Constructing summary statistics for ABC. *JRSS-B*, 2012.
- Khoo et al. Local Fisher score matching. 2026.
- Jiang et al. [Surrogate-score architecture comparator — complete citation before circulation.] 2025.
- C. Varin, N. Reid, D. Firth. An overview of composite likelihood methods. *Statist. Sinica*, 2011.
- V. Chernozhukov and H. Hong. An MCMC approach to classical estimation. *J. Econometrics*, 2003.

Claim	Experiment
Thm. 8 (Q1): nonlinear > linear	Elliptical AR-scatter field (Ex. 1), closed-form g_0 : plot $e_\gamma^\top n^{-1}\{I - G_\star^{\text{lin}}\}e_\gamma$ (flat, positive) and $e_\gamma^\top n^{-1}\{I - G^{\text{nl}}\}e_\gamma$ ($\rightarrow 0$) vs. n . Sweep the generator toward Gaussian: gap must close. <i>Report the scatter direction, not ν</i> (saturation, Prop. 12).
Thm. 14 (Q2): pre-agg. > outer	Block latents (Ex. 3). Three panels: (a) $\text{RE}_C \approx \text{RE}_A$; (b) deficit flat in m at fixed J while C's edge over A shrinks; (c) deficit increasing in J or d . Blockwise nCS $\rightarrow 1$; global pooling does not. <i>Headline figure</i> .
Prop. 12: honest concession	Global mixture / global elliptical (Ex. 2): show $\text{RE}_C \rightarrow 1$ and $I_n(\nu) = O(1)$. Conceding this pre-empts the sharpest referee objection.
Thm. 8(ii) + Prop. 6: two negative controls	(a) Gaussian banded-precision field (Ex. 4); (b) any first-order Markov model with pairwise+marginal library. Gap is <i>exactly</i> zero in both; nonlinear must match , not beat. Beating either falsifies the pipeline.
Principle 1: granularity, not flexibility	Delay- k threshold AR (Ex. 5): pointwise $\phi \approx +0.01$, windowed ϕ at matched granularity $\approx +0.09$, triple-margin linear CS exact.
Thm. 10(c): geometric window law	HMM / stochastic volatility (Ex. 6): $1 - \text{RE}$ vs. window width w on a log scale, slope $\approx 2 \log \rho$; sweep persistence. Supplies a rate, not a dichotomy.
Prop. 16: screening works	Estimate $\text{Var}(e)/\text{Var}(g)$ from pilot sims before training; compare against realized nCS gain across all models above. A working a-priori diagnostic is a contribution in itself.
Credit attribution	$W^\top T_\phi$ (linear readout, learned features) vs. $S_\psi(T_\phi)$ (nonlinear readout): isolates aggregator from features.
Baselines are not merely under-fed	Khoo / Jiang architectures <i>given the sub-score features as inputs</i> , matched simulation budget and capacity. If the gap closes, the contribution is the features, not the aggregation — run this before submission.
Cor. 23: degeneracy	Summary = $S_{\psi,\phi}(X, \hat{\beta})$ alone vs. pair vs. $\hat{\theta}_1$. Expect pair $\geq \hat{\theta}_1 \gg$ score-alone; first gap shrinks in n , last never closes.
Prop. 29 + 31: validity vs. efficiency	2×2 : {Alg. 6, Alg. 7} $\times$ {good pilot, biased pilot}. Plus: <i>inconsistent</i> pilot with tube centered on its biased law $\Rightarrow$ calibrated but wider posteriors. No classical method produces this plot.
Prop. 33: sandwich validity	Coverage of $\hat{\Sigma}$ -based intervals; SBC of the NPE posterior; agreement between NPE spread and $\hat{\Sigma}$.
Deployment	Real heavy-tailed time series. Cannot evidence the gap (no ground truth); report calibration via simulation from $\hat{\theta}$. Say so explicitly in the text.